MILITARY SPECIFICATION SHEET

ELECTRON TUBE, PHOTOMULTIPLIER

TYPE 1P21

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: 9-stage, side-on, S4 spectral response

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ebb	Eb-Edy9	Light source	Light	Ib	ib	TA	Rp	Pp	Cathode area	Shock	Alt
Unit:	Vdc	Vdc	---	ℓm	mAdc	ma	°C	Ohms	W	Lighted sq cm	G	ft
Maximum:	1,250 (Note 20)	250 (Note 20)	---	---	0.1	1.0	75	Note 3	0.25	---	200	40,000
TEST CONDITIONS:	1,000 (Note 1)	100 (Note 2)	LS1	10^{-5}	---	---	---	10,000	---	0.8 (Note 4)	---	---

GENERAL:

Qualification - Required

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MIL-STD-750	Test	Conditions	Symbol	Limits		Unit
				Min	Max	
	Qualification inspection					
1031	Low-frequency vibration	No voltages; t 60 minutes	---	---	---	---
1331	Direct-interelectrode capacitance	See note 5	C	2.0	6.0	pF
1331	Direct-interelectrode capacitance	See note 6	C	3.5	9.5	pF
---	Low-pressure breakdown	Ebb 1,250 Vdc; Eb to Edy9 125 Vdc; voltages for all other stages 125 Vdc per stage; Rp 1 Meg (see notes 7, 8, and 9)	---	---	---	---
5303	Spectral response identification	S4	---	---	---	---
---	Plastic base material	See note 13	---	---	---	---
	Quality conformance inspection, part 1					
---	Cathode current	Light 0.01 ℓm (see notes 10, 11, 12, and 19)	Ik	0.20	---	μAdc
5301	Anode current	See notes 10, 12, and 19	Ib	400	---	μAdc
5301	Dark current	Light 0 (see note 19)	Lik	---	2.5	μAdc
5301	Dark current (1)	Light 0 (see note 19)	Lib	---	0.05	μAdc
5301	Dark current (2)	Light - 0; Ebb 1,250 Vdc (see note 19)	Lib	---	1.0	μAdc
5301	Dark current (3)	Ebb 500 Vdc; Eb to Edy9 50 Vdc; Rp 10,000 ohms (see note 19)	Lib	---	0.006	μAdc
5301	Dark current (4)	Light 0 (see notes 14 and 19)	Lib	---	0.01	μAdc
5306	Signal-to-noise ratio	Light 10^{-8} ℓm; Rp = 0.1 Meg (see notes 15 and 19)	S/N	48	---	dB
	Quality conformance inspection, part 2					
1101	Secureness of base, cap, or insert	See note 16	---	---	---	---
1041	Shock	2000 ± 10; (see note 17)	---	---	---	---
(G) 1111	Base pin solder depth		---	---	---	---
1105	Permanence of marking		---	---	---	---
	Quality conformance inspection, part 3					
---	Life-test provisions	Group C; t = 500 hours (see note 4)	---	---	---	---
---	Life-test end points	Anode current (see notes 10 and 18)	---	---	---	---

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NOTES:

1. Overall anode to cathode dc supply voltage.

2. Anode to 9th dynode dc voltage.

3. An anode load resistance of at least 10,000 ohms is recommended for a protective resistance.

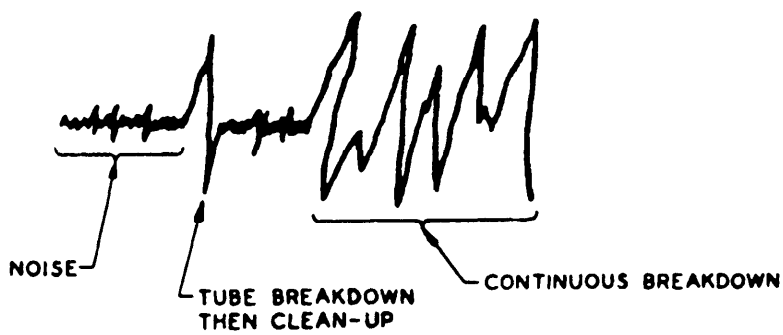
4. Ebb = 1,000 V adjust luminous flux at cathode initially to obtain Ib - 0.1 mAdc. Color temperature shall be greater than 2,000°K.

5. Capacitance between anode and dy9, all other elements grounded.

6. Capacitance between anode and all other elements.

7. Test circuit and test equipment requirements:

A bleeder network to provide equal voltage per stage; 8-megohm series resistance for cathode, and all dynodes excluding the last dynode (9th); 1-megohm series resistance for last dynode (9th) and anode; a regulated (0.1 percent) power supply with low ripple (0.05 volts peak-to-peak maximum) as a voltage source; a sensitive dc microammeter to determine anode current; an oscilloscope, with a low-capacity divider, is connected across the anode load resistance (1 megohm) to monitor noise and breakdown; base pins of photomultiplier under test connected so that the degradation of breakdown characteristic is minimized; a vacuum pump, low-pressure chamber and manometer; a light source with an adjustable iris and sandblasted glass. Mask tube with lightproof material. The light source should be located so that light from it is striking the cathode of the photomultiplier. (Light from the light source should be the only light striking the cathode.) The opening of the iris is varied to obtain approximately 10 microamperes of anode current. Observe the tube noise signal as displayed by the oscilloscope. Reduce pressure of vacuum chamber to 130 mmHg. The tube should be subjected to this pressure for a period of 2 minutes. During pressure reduction, the tube noise signal, as displayed by the oscilloscope, should be observed to determine tube breakdown. When breakdown occurs, the signal being monitored by the oscilloscope will resemble the following:



When continuous breakdown occurs, a change in anode current will result.

8. Vary light to obtain approximately 10 microamperes of anode current.

9. The tube shall operate without indications of breakdown in an evacuated chamber in which the pressure does not exceed 130 mmHg absolute.

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NOTES: -Continued

10. Adjust tube position to give maximum sensitivity.

11. Apply 100 volts dc between cathode and all other electrodes. The pins shall be tied together.

12. The light during test, except during life test, shall be incident upon an aperture approximately 0.8 inch by 0.2 inch, located to center on the cathode.

13. With the qualification test report, the tube manufacturer shall certify that the base used has been fabricated from an insulating material qualified under MIL-M-14 for type CFG. Bases previously qualified to method 1216 are acceptable.

14. The overall anode to cathode voltage in this test shall be determined as follows:

With a light flux of 10^{-5} lumens, the supply voltage, epp, shall be varied until the anode current is 180 μA. With the light flux set at 0, this voltage now serves as the test voltage.

15. Prior to testing, the tubes shall be kept in the dark at an ambient temperature of 70° ±5° F for not less than 1 hour. The light spot on the photocathode, based on a projected dimension perpendicular to beam axis, shall be 1/8 to 3/16 inch (3.18 to 4.76 mm).

16. Immersion in water shall be omitted and tubes shall be tested dry. Torque shall be applied with 20.0 in.-lb minimum.

17. After the shock test is performed, the tube shall meet the requirements of all the tests designated as quality conformance inspection tests, part 1.

18. Anode current limit is 40 μAdc minimum or 1/20 of initial Ib, whichever occurs first.

19. This test to be performed at the conclusion of the holding period.

20. An ac supply voltage may be used having a peak value equal to this maximum value.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army -
Navy -
Air Force - 11,
DSA - ES

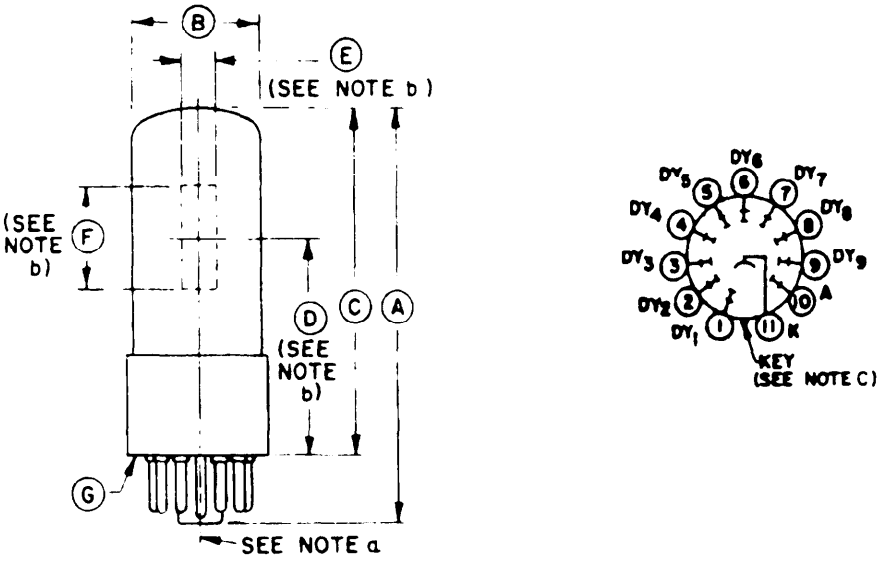
User activities:
Army - MU, WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2425-60)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
C		3.125 (79.38)
E	.312 (7.92)	
F	.937 (23.80)	
G	Base: B11-88 (see note c)	
Quality conformance inspection, part 2		
A		3.687 (93.65)
B		1.187 (30.15)
D	1.843 (46.81)	2.031 (51.59)

NOTES:
a. Centerline of bulb shall not deviate more than 2° in any direction from the perpendicular erected at center of bottom of base.
b. Dimensions D, E, and F refer to the usable cathode area.
c. Photosensitive surface of cathode shall be oriented with respect to base guide key.

FIGURE 1. Outline drawing of electron tube type 1P21.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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